One Stream, Many Windows: Soft Targets for Consolidating Fast and Slow Conversions

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Abstract. Conversion pipelines in advertising are split by speed: an upper-funnel onsite conversion is observable within minutes of the click, while a down-funnel offsite conversion takes a day or more. Each objective defines an *attribution window*, the elapsed time within which a conversion counts. Conventionally a training pipeline's join delay is the same as conversion attribution window. Tying the two together forces every objective onto its own pipeline, each as stale as the window it waited for. We ask whether they can be set independently. Writing records sooner makes every conversion still in flight become a negative, and a streaming pipeline that trains each click at most once cannot revise them afterwards. Yet most published corrections do exactly that: of 9 methods we reproduce, 7 write more than one record per click, up to 4.00. On the Criteo Delayed Feedback Dataset we compare three targets across 6,900 runs, five served capacities, three join delays, two attribution windows and two teacher architectures. Writing the record early and replacing only the negatives with a teacher's estimate beats waiting for the matured label in 28 of 30 configurations, by 1.97% at seven days and 0.57% at one, which is 74% and 69% of what a label-peeking oracle attains. It is the strongest method obeying the one-record rule.

Keywords: Conversion rate prediction · Delayed feedback · Online advertising · Knowledge distillation · Streaming training pipelines.

1 Introduction

Advertising systems predict several types of conversion at once, and those types differ sharply in how long they take to observe. A website view or an add-to-cart is recorded within minutes of the click; a purchase, an app install or a submitted lead may take a day or a week. Each objective is defined with an *attribution window*, the elapsed time within which a conversion is credited to the click, and a production system maintains several windows at once. Conventionally that window also determines the *join delay*, the age at which the click's training record is written to the stream: the pipeline waits until the label is knowable, then emits the record. This conversion has two disadvantages: A model for a slow objective trains only on data at least an attribution window old, so it never sees fresh data during online training. And a system that ranks several objectives jointly must either maintain one pipeline per objective or delay all of them until

the slowest window closes, imposing the staleness of the slowest funnel on the fastest.

Neither those two disadvantages are intrinsic to conversion modeling. The attribution window is fixed by the business definition of a conversion; but the join delay is a pipeline choice. We ask what happens when the two are set independently, and specifically whether an objective accountable to a seven-day window can be trained from records joined within minutes.

The obstacle is that an early record is wrong in a known direction: every conversion still in flight when the record is written is recorded as a negative. A substantial literature corrects this censoring bias [1,11,10,2,4], but most corrections revise the record afterwards, by duplicating the click when its outcome lands, by relabelling it, or by adding prediction heads at further horizons. A streaming pipeline that appends each record once and trains on it once can do none of these. We state that restriction as a *single-visit* constraint and measure which published methods satisfy it.

Our idea is this censoring bias can be mitigated by the prediction from a model that has seen trained on the matured outcomes in the conventional way, i.e. *a teacher model*. Such a teacher model is often readily available in production, where large ranking models are already trained across objectives on fully matured data [8]. The contributions of this work are: (1) a separation of the join delay from the attribution window, with the causality and single-visit rules a deployable pipeline must satisfy, and an audit of 9 published methods against them, of which 7 violate single-visit by writing up to 4.00 records per click; (2) evidence that writing the record early and replacing only its observed negatives with teacher model predictions beats waiting for the matured label, by more as the funnel slows, and is the strongest target obeying the single-visit rule; (3) the finding that at small served capacities the hard label can be dropped altogether, since a training pipeline consisting only of the teacher's estimations as labels can beat the conventional training pipeline with delayed join same as attribution windows, which can benefit early stage, such as retrieval, where model complexity is low.

2 Problem Setting and Proposed Approach

A click i occurs at time C_i with features x_i and, if it converts, does so after a delay $D_i > 0$, with $D_i = \infty$ when it never converts. The attribution window v belongs to the objective and fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$. The join delay o belongs to the pipeline and fixes when the record is written, so the label available at that moment is

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

Setting $o = v$ reproduces the conventional pipeline. Setting $o \ll v$ is what we want in order to consolidating fast and slow conversions into the same training pipeline.

Two rules a deployable pipeline must satisfy. **Rule 1, causality.** A record's label must be computable at the moment the record is written, and must not change afterwards. **Rule 2, single-visit.** Let n_i be the number of times click i enters the served model's loss over its lifetime. Rule 2 requires $\max_i n_i \leq 1$. Rule 2 is a practical rule used in the production systems.

Which published methods comply. We reproduced 9 published methods from scratch under a single protocol and counted the records each one's served model consumes per distinct click. 7 write more than one. At $v = 1\,\mathrm{d}$ the counts are FTP [5] 4.00, MISS [6] 3.05, PI [12] 2.00, DDFM [3] 1.89, DFM [1] 1.33, DEFUSE [2] 1.03, ES-DFM [10] 1.03 . Only FSIW, ULC satisfy Rule 2. Of the two, FSIW corrects the loss and leaves the target itself untouched, while ULC changes the target, which makes it our closest comparator.

Four targets. We studied four types of pipeline targets: FRESH writes at $\Delta = o$ with $\tilde{y}_i = Y_i^{(o)}$, filling the deficit with nothing. WAIT writes at $\Delta = v$ with $\tilde{y}_i = Y_i^{(v)}$, which is the conventional method and our comparison baseline. CORRECT-DISTILL writes at $\Delta = o$, keeps the observed labels and softens only the observed negatives with a teacher model's prediction.

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \;\Big|\; Y^{(o)}{=}0,\, x\Big)\,. \tag{2}$$

An observed positive stays exactly 1, and only the uncertain mass moves. This is the functional form used by ULC [9]; the two differ in how w^T is estimated. DISTILL-ONLY goes further and replaces both all observed labels with a teacher model's prediction. A teacher model in both DISTILL-ONLY and CORRECT-DISTILL is trained with the conventional WAIT method. Finally, ORACLE writes at $\Delta = o$ but labels the record $Y_i^{(v)}$. It violates Rule 1 and is not deployable. We include it because it bounds what any causal correction could recover.

3 Experiments and Results

3.1 Experimental Setup

Data and protocol. We use the public Criteo conversion logs released with [1]. Clicks span 60 days. Days $[0, 30)$ initialise the model causally and days $[30, 60)$ form the streaming period, divided into 719 hourly rounds yielding 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a seven-point grid on a held-out slice of the initialisation period, before any streaming evaluation, then pinned across seeds. The teacher's rate is fixed and decoupled from the student's.

Model architecture. In the model arch backbone, the eight integer fields enter a dense arm as $\log(1+x)$ paired with an observed-mask bit, normalised and passed through a two-layer perceptron to a projection vector. The nine categorical fields

are hashed per field and embedded, one identifier per field, giving a token per field. Tokens and the dense projection are then consumed by a stack of Wukong interaction layers. The final token set is flattened to a single logit for final output.

Scaling complexity of the served model. The five rungs M_0 to M_4 move three dials together rather than one: the interaction trunk is taken from successive rungs of a compute-efficient frontier, the hash width per categorical field grows $36 \to 256 \to 1{,}280 \to 16{,}384 \to 131{,}072$, and the embedding dimension widens $4 \to 4 \to 8 \to 8 \to 8$. The result is roughly an order of magnitude per step, 1,890 to 9,580,477 parameters and 0.001 to 0.286 MFLOPs per inference.

Different setups and metric. Two attribution windows ($v \in \{1, 7\}$ days), three join delays ($o \in \{5, 30, 90\}$ minutes), serving models with five different complexities (M_0 to M_4), and two different teacher models are used. For the five different complexities, the hash width per categorical field grows $36 \to 256 \to 1{,}280 \to 16{,}384 \to 131{,}072$, and the embedding dimension widens $4 \to 4 \to 8 \to 8 \to 8$. The result is roughly an order of magnitude per step, 1,890 to 9,580,477 parameters and 0.001 to 0.286 MFLOPs per inference. For the two teacher models, *Teacher 1* places 8-dimensional embedding tables under a two-layer perceptron, giving 9,456,897 parameters at 0.039 MFLOPs. *Teacher 2* places 48-dimensional tables under a deep interaction trunk, giving 57,272,515 parameters at 1.593 MFLOPs. Running both is what lets us report the method's sensitivity to the choice of teacher rather than to one arbitrary configuration of it. For each step, we ran 5 seeds for statistic analysis. We report normalised entropy and calibration against the matured $Y^{(v)}$ for all methods.

3.2 Findings

A short join delay beats waiting, and by more on slow funnels. Writing the record 2,016 times sooner than the seven-day window it answers to reduces NE in all 15 seven-day cells, 15 of them surviving Holm–Bonferroni correction, ranging from 0.48% to 2.68% with a mean of 1.97%. At one day the same substitution gains 0.57%, positive in 13 of 15 cells and significant in 10. The direction is the informative part: the slower the funnel, the more a short join delay is worth, because WAIT at $v = 7\,\mathrm{d}$ trains on week-old data and no amount of label accuracy compensates for that. Both ends of the interval are measured. ORACLE gains 2.68% and 0.83%, so CORRECT-DISTILL recovers 74% and 69% of what any causal correction could attain, while FRESH loses 20.1% and 11.8%. The deficit the teacher fills is an order of magnitude larger than the gains above it.

At small capacity the hard label can be dropped entirely. DISTILL-ONLY carries no observed label at all. At the two smallest capacities it beats WAIT in all 12 of 12 cells, by 0.60% at $v = 1\,\mathrm{d}$ and 0.85% at $v = 7\,\mathrm{d}$, which suits retrieval and other early-funnel stages: the record reduces to features and one number, and emission needs no join against the outcome. It does not carry to the larger rungs, losing 0.51% at $v = 1\,\mathrm{d}$ across the three largest capacities and staying

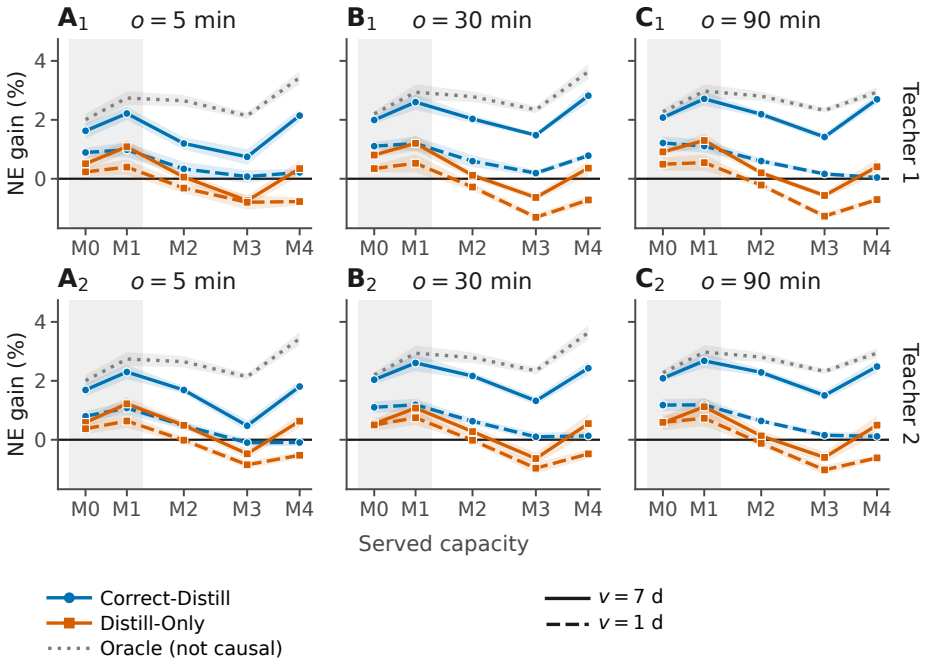


Fig. 1. Relative NE reduction against Wait as a function of served capacity. Columns are the join delay, (A) $o = 5$, (B) $o = 30$ and (C) $o = 90$ minutes; rows are the teacher the targets were distilled from, teacher 1 in A_1–C_1 and teacher 2 in A_2–C_2. Each row is read against its own Wait runs, and the near-identical rows are the teacher-robustness result. Above the zero line beats the conventional per-objective pipeline. Colour denotes the target and line style the attribution window; Oracle, which labels an early record with the matured outcome and is therefore not causal, is drawn at $v = 7\,\mathrm{d}$ to bound each panel. Shading on a curve is the standard deviation of the per-seed paired differences; the grey vertical band marks the two smallest capacities, the regime in which Distill-Only needs no hard label.

ahead by only 0.10% at $v = 7\,\mathrm{d}$, crossing over between M_1 and M_2 at every join delay, plausibly because a soft target acts as a regulariser that a larger model no longer needs [7]. Where capacity is ample the observed positive is what pays: Correct-Distill beats Distill-Only in all 30 cells and Wait in 28 of 30. The 2 exceptions are both at $v = 1\,\mathrm{d}$ and $o = 5$ minutes, at rungs M3, M4, the worse by 0.097%, and neither survives correction; that is where the teacher has least to add, since 0.355 of the conversions are already visible.

Comparison with published methods at a production-like operating point. Table 1 fixes one cell, $v = 1\,\mathrm{d}$ and $o = 90$ minutes, the point a large-scale system sits at, and reports the two ends of the capacity ladder rather than a mean over them.

Three readings follow. First, the rungs disagree. At M_0 correcting the early write is worth $+1.19 \pm 0.12\%$ and takes 97% of the $+1.23\%$ Oracle headroom,

Table 1. All reproduced methods at $v = 1\,\mathrm{d}$ and $o = 90$ minutes, the operating point closest to a large-scale production setting, at the smallest and largest served capacities. NE and calibration error are absolute and lower is better; ΔNE is the per-seed paired reduction against WAIT in percent, higher is better, reported with the half-width of its 95% t interval. The replicate is a seed, five per arm and ten for WAIT, and a (teacher, seed) pair for the two arms in bold, which are ours and ran under both teachers. ‡ORACLE writes one record but labels it with the matured outcome, so it violates Rule 1 and is an upper reference rather than a method.

	M_0 (smallest)			M_4 (largest)		
Method	NE	Cal.	ΔNE (%)	NE	Cal.	ΔNE (%)
Single visited: yes						
WAIT	0.7865	0.0578	ref.	0.7209	0.0737	ref.
FSIW [11]	0.7785	0.0553	$+1.03 \pm 0.26$	0.7173	0.0781	$+0.51 \pm 0.09$
ORACLE‡	0.7769	0.0489	$+1.23 \pm 0.28$	0.7194	0.0617	$+0.22 \pm 0.08$
CORRECT-DISTILL	0.7772	0.0555	$+1.19 \pm 0.12$	0.7203	0.0676	$+0.09 \pm 0.06$
ULC [9]	0.7773	0.0529	$+1.18 \pm 0.22$	0.7207	0.0707	$+0.04 \pm 0.06$
DISTILL-ONLY	0.7821	0.0672	$+0.56 \pm 0.14$	0.7256	0.0782	-0.65 ± 0.06
FRESH	0.7929	0.1855	-0.80 ± 0.25	0.7345	0.1813	-1.88 ± 0.10
Single visited: no						
DDFM [3]	0.7902	0.0632	-0.46 ± 0.31	0.7168	0.0572	$+0.57 \pm 0.09$
PI [12]	0.7806	0.0692	$+0.76 \pm 0.43$	0.7181	0.0811	$+0.40 \pm 0.03$
FTP [5]	0.7786	0.0515	$+1.02 \pm 0.18$	0.7187	0.0636	$+0.31 \pm 0.16$
DEFUSE [2]	0.7789	0.0681	$+0.98 \pm 0.30$	0.7217	0.0856	-0.10 ± 0.07
ES-DFM [10]	0.7791	0.0690	$+0.96 \pm 0.31$	0.7219	0.0869	-0.13 ± 0.08
DFM [1]	0.7569	0.0612	$+3.78 \pm 0.19$	0.7236	0.0584	-0.37 ± 0.09
MISS [6]	0.7830	0.0826	$+0.45 \pm 0.24$	0.7268	0.1229	-0.82 ± 0.23

whereas at M_4 that headroom is itself only $+0.22\%$ and we take 42%, because 0.803 of this cell's conversions are already visible when the record is written. Second, ORACLE bounds only the methods that obey Rule 2: DFM reaches $+3.78\%$ at M_0, far above it, because duplicating a click supplies training signal that matters most when the model has least capacity to extract it. Third, among methods a streaming pipeline can run we lead at M_0, where ULC [9] reaches $+1.18\%$ inside our interval, but not at M_4, where FSIW [11] reaches $+0.51 \pm 0.09\%$ by correcting the loss rather than the target, having averaged -0.45% over this window's 15 cells; across the slow window we lead ULC by $+0.13$ pp in 12 of 15 cells. The gain also costs nothing in calibration: CORRECT-DISTILL is better calibrated than WAIT at both rungs, while FRESH degrades calibration error by 3.2 and 2.5 times.

Robustness to the teacher. Substituting teacher 1 for teacher 2 changes little. CORRECT-DISTILL gains 2.00% against 1.97% at seven days and 0.63% against 0.57% at one, and over all 30 cells the two differ by -0.043 pp for CORRECT-DISTILL and $+0.114$ pp for DISTILL-ONLY. Spending 4.7 times more teacher compute at fixed parameters stays inside the same margin, and the default

teacher learning rate beats one ten times smaller in every cell tested, so this describes a converged teacher rather than an untuned one. Freezing the teacher after initialisation, so that it ships as a static file rather than a second training job, costs 0.08 and 0.27 pp and still beats WAIT in 27 of 30 cells. The method therefore does not depend on a particular teacher, nor on keeping one running. That insensitivity is partly a property of the corpus: Criteo carries eight integer and nine hashed categorical fields with no creative, content or user-history signal, so $P(Y^{(v)}{=}1 \mid Y^{(o)}{=}0, x)$ is close to exhausted by a modest architecture. A production corrector would see far richer features, and a cross-objective foundation model trained on fully matured outcomes [8] is the artefact positioned to supply it. We expect that to widen the gains reported here, but we did not measure it.

4 Concluding Remarks

An objective's attribution window is fixed by the business definition of a conversion; the join delay its pipeline adopts is not. Setting the two independently improves accuracy rather than degrading it, recovering 74% of what a label-peeking oracle attains at seven days and helping most where the funnel is slowest. Which target to write depends on the capacity served: above the smallest rungs the record should keep its observed positive, while at small capacity the label can be dropped and the teacher's estimate carried alone.

The consequence is for training several ranking objectives in one model, an active industrial direction [8] whose first obstacle is a pipeline obstacle rather than a modelling one. Under WAIT, objective k's record exists only at $C_i + v_k$, so a shared append-only stream must either write K records per click, violating Rule 2 K times over, or delay every objective until $\max_k v_k$. A single stream at one short join delay instead carries $\tilde{y}_{i,k} = Y^{(o)}_{i,k} + (1 - Y^{(o)}_{i,k})\, w_k^T(x_i)$ for every objective at once, because w_k^T is a function the pipeline evaluates rather than an outcome it awaits. ULC reaches comparable accuracy, so the choice between them is architectural: ULC fits each w_k inside the served model's training loop, whereas a teacher places each upstream with a single number written onto a record as its only interface. Section 3 shows that upstream model need not be retrained while the student trains, so consolidation adds K static files to the build rather than K running training jobs.

Two limits bound the claim. We ran one objective at a time, so we have no evidence on interference between objectives sharing a trunk, which is the central risk in the target above, and one corpus with one chronological split measures precision rather than generalisation. The experiment this makes worth running is K prediction heads, K attribution windows, one stream, and the teachers placed upstream.

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